SMOOTHNESS AT A POINT OF A MORPHISM OF ALGEBRAIC SPACES IS EQUIVALENT TO LIFTING ALONG SMALL EXTENSIONS THERE

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Editorial note. Statement span is lines 5095-5117 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 5119-5152, the complete author proof (32 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Smoothness at a point, small extensions of local rings, and the schemes-side Artinian lifting criterion; the reduction is retained.

Source reference. Original lemma algebra-definition-small-extension: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5152–5152. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-etale-formally-etale: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3641–3649. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-smooth-formally-smooth: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4587–4595. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-morphisms-lemma-lifting-along-artinian-at-point: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5152–5152. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $x \in |X|$. Assume that Y is locally Noetherian and f locally of finite type. The following are equivalent:*

(1) *f is smooth at x,*

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

(2) *for every solid commutative diagram*

$$\begin{array}{ccc} X & \xleftarrow{\alpha} & \mathrm{Spec}(B) \\ {\scriptstyle f}\downarrow & \nearrow & \downarrow{\scriptstyle i} \\ Y & \xleftarrow{\beta} & \mathrm{Spec}(B') \end{array}$$

where $B' \to B$ is a surjection of local rings with $\mathrm{Ker}(B' \to B)$ of square zero, and α mapping the closed point of $\mathrm{Spec}(B)$ to x there exists a dotted arrow making the diagram commute, and

(3) *same as in (2) but with $B' \to B$ ranging over small extensions (see Algebra, Definition algebra-definition-small-extension).*

Proof. Condition (1) means there is an open subspace $X' \subset X$ such that $X' \to Y$ is smooth. Hence (1) implies conditions (2) and (3) by Lemma lemma-smooth-formally-smooth. Condition (2) implies condition (3) trivially. Assume (3). Choose a commutative diagram

$$\begin{array}{ccc} X & \longleftarrow & U \\ \downarrow & & \downarrow \\ Y & \longleftarrow & V \end{array}$$

with U and V affine, horizontal arrows étale and such that there is a point $u \in U$ mapping to x. Next, consider a diagram

$$\begin{array}{ccccc} X & \longleftarrow & U & \xleftarrow{\alpha} & \mathrm{Spec}(B) \\ \downarrow & & \downarrow & & \downarrow{\scriptstyle i} \\ Y & \longleftarrow & V & \xleftarrow{\beta} & \mathrm{Spec}(B') \end{array}$$

as in (3) but for $u \in U \to V$. Let $\gamma : \mathrm{Spec}(B') \to X$ be the arrow we get from our assumption that (3) holds for X. Because $U \to X$ is étale and hence formally étale (Lemma lemma-etale-formally-etale) the morphism γ has a unique lift to U compatible with α. Then because $V \to Y$ is étale hence formally étale this lift is compatible with β. Hence (3) holds for $u \in U \to V$ and we conclude that $U \to V$ is smooth at u by More on Morphisms, Lemma more-morphisms-lemma-lifting-along-artinian-at-point. This proves that $X \to Y$ is smooth at x, thereby finishing the proof. □